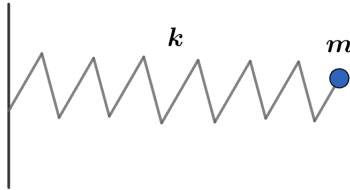


2014 F=ma Exam: Problem 11

Kevin S. Huang

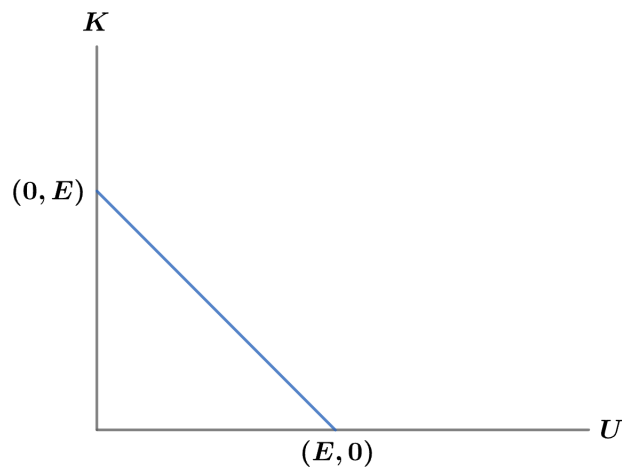


By conservation of energy, we have

$$E = K + U$$

$$K(U) = -U + E$$

which is a line with slope -1 and intercept E .



Since $K = \frac{1}{2}mv^2 \geq 0$ and $U = \frac{1}{2}kx^2 \geq 0$, the plot only exists in the first quadrant. Thus, the answer is $\boxed{E}$.